

HA delivered and monitored

Day 7 - Morning

Running scenario: Thara Hospitals patient portal pilot

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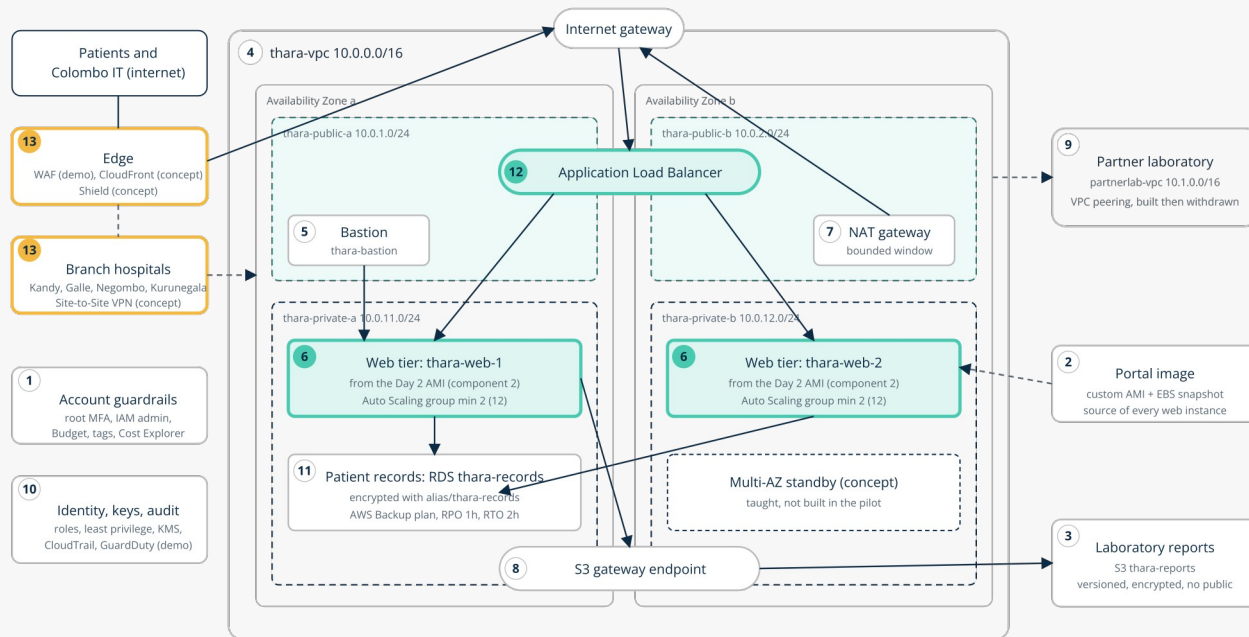
Today



- **Availability as arithmetic:** a percentage, read as minutes
- **Making it heal itself:** a template, a group, a load balancer, an alarm
- **The edge, demonstrated:** a control in front of the load balancer
- **The bill, read properly:** Cost Explorer against your ledger

Then you build component 12, and delete it before the break.

The Thara architecture



TODAY

12 Load balancer and Auto Scaling group

13 Edge and branches

6 Portal web tier

KEY

Built today

Built earlier

Demonstrated

CONCEPT 1

Availability as arithmetic

One flood reached everything. Two zones and two servers are the fix.

25 min

Read a target as minutes in a month

438

minutes at 99%
a person can meet it

44

minutes at 99.9%
recovery must start itself

4

minutes at 99.99%
nothing may need recovering

Design	The sum	Downtime in a month
One part at 99.9%	0.999	about 44 minutes
Three parts in a chain	$0.999 \times 0.999 \times 0.999$	about 131 minutes
One server at 99%	0.99	about 438 minutes
Two servers side by side	$1 - (0.01 \times 0.01)$	about 4 minutes

Fault tolerance or high availability?

Fault tolerance

When something fails

No interruption at all.

What it takes

Spare capacity already running and carrying the load.

High availability

When something fails

A short interruption, then it recovers by itself.

What it takes

More than one of everything, in more than one place.

The pilot buys **high availability**.

Disaster recovery: a ladder



- **Backup and restore:** copies of the data, nothing running. Day 6
- **Pilot light:** data kept current, servers defined and switched off
- **Warm standby:** a small working copy, made larger
- Each rung costs more and recovers sooner. Multi-site is a name only

The RPO and RTO say how high to climb. The budget says how high you can.

Scaling: up, or out

- **Vertical:** one larger server. Stops at the largest you can buy
- **Horizontal:** more servers of one size. Stops where they are not identical

YOU ALREADY KNOW THIS

Redundant supervisors in a chassis versus a stack of switches.

T**THARA HOSPITALS**

The monsoon flood was a single point of failure. Two AZs and two instances are the fix the pilot can afford.

CONCEPT 2



Making it heal itself

Something must share the requests, notice a failure and start another.

25 min



CHIEF FINANCIAL OFFICER

OPD peaks and dengue season. I pay for the third instance only when it is needed.

The Chief Financial Officer, on why the group has a maximum.

Template, group, health check

- **Launch template**: what a web server is
- **Auto Scaling group**: how many there must be
- Minimum, **desired**, maximum: fewer than desired, start one
- Health checks come from the load balancer

YOU ALREADY KNOW THIS

A hardware load balancer health-checking a server pool.

COMMON MISTAKE

Auto Scaling replaces the load balancer.

The ALB is separate and needs to be created.

Growing needs a rule

- A **scaling policy** acts when a CloudWatch alarm fires
- A **cooldown** follows, so one busy minute adds one server
- A one-minute alarm needs detailed monitoring
- Metrics, alarms, logs. **SNS** carries the notification

COMMON MISTAKE

The ALB is in a subnet.

It spans subnets across AZs.

CONCEPT 3



The edge, demonstrated

A patient portal on the public internet is probed within hours.

15 min

WAF: a web ACL on the load balancer

- A **web ACL** reads each web request and allows or blocks it
- Managed rule groups cover the common attacks
- Demonstrated, not built: the point is **placement**, not configuration
- CloudFront and Shield, from Day 4, are the next phase



THARA HOSPITALS

A patient portal on the public internet will be probed within hours of going live.

Demonstration: WAF on the load balancer



- 
1. Create a web ACL with the AWS managed core rule group and attach it to a load balancer
 2. Send a request with a SQL-injection-shaped query string. It is blocked
 3. Show the sampled request in the WAF console
 4. Delete the web ACL at the end of the session

CONCEPT 4

The bill, read properly

Your ledger and Cost Explorer describe the same weeks.

15 min

Two ways to cut the bill

- Group by **Service**: where the money went
- Group by the `Owner` tag: whose money it was
- A tag groups costs only once it is activated for billing
- Ledger and Cost Explorer differ: **timing**, and **credits**

A difference that neither explains means something was left running.



The CFO's actual against forecast.

HA delivered and monitored



You build:

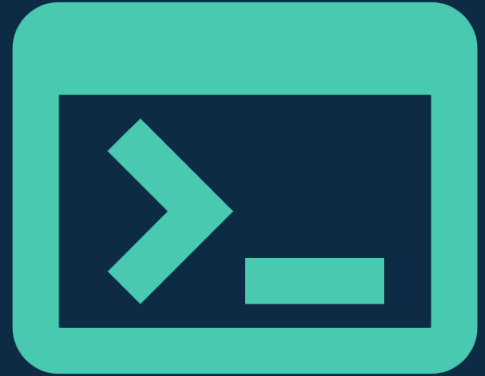
12

13

6

Estimated credit use: about USD 1.50

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SEGMENT 1 - 20 MIN



Launch template and target group

Cost: nil

Check, then say what a web server is

1

Region and user. Service Quotas: read the vCPU limit for standard instances. Start `thara-bastion` only, and set its SSH source to My IP.

Expected: A limit of 8 or more. `thara-web-1` stays stopped.

2

Launch template `thara-web-lt`: `thara-portal-v1`, `t3.micro`, `thara-key`, `thara-web-sg`, `thara-web-role`, tags with `Day=7`.

Expected: `thara-web-lt` at version 1. No subnet in it.

Enable detailed monitoring, and paste the user data from the lab sheet.

The list the load balancer sends to



3

Target group `thara-web-tg`: instances, HTTP 80, `thara-vpc`, health check path `/`. Register nothing.

Expected: No targets yet. Every 30 seconds; 2 failures; 5 successes.

SEGMENT 2 - 20 MIN



ALB across both public subnets

Cost: about USD 0.025 per hour plus LCU

Create the load balancer, and write the time

4

Security group `thara-alb-sg`: HTTP from anywhere. Then `thara-alb`: internet-facing, `thara-public-a` and `thara-public-b`, listener 80 to `thara-web-tg`.

Expected: Active in a few minutes, with a DNS name. Copy it.

4

In `thara-web-sg`, change the HTTP source from anywhere to `thara-alb-sg`.

Expected: Web requests now come from the load balancer only.

This bills by the hour. It is deleted before the break.

SEGMENT 3 - 20 MIN



ASG min 2 across both private subnets

Cost: two to three t3.micro

Say how many there must be



5

Auto Scaling group `thara-web-asg` from `thara-web-lt`, in `thara-private-a` and `thara-private-b`, attached to `thara-web-tg`. Minimum 2, desired 2, maximum 3.

Expected: Two instances, one in each zone. Healthy after about two and a half minutes.

6

Browse to the DNS name and reload a dozen times.

Expected: The pilot page, served by two instance ids.

Turn on Elastic Load Balancing health checks.

SEGMENT 4 - 25 MIN



Alarm, load, scale-out, replacement, SNS

Cost: nil

A topic, an alarm, a policy



7

SNS topic `thara-alerts`, with your email subscription. Confirm it from your inbox.

Expected: The subscription reads Confirmed.

8

Alarm `thara-web-asg`: the group's average CPU above 50 for 2 periods of 1 minute, notifying `thara-alerts`. Then a simple scaling policy: add 1, wait 300 seconds.

Expected: The alarm reads OK. The group shows one policy.

Load, then destroy

9

From the bastion, connect to each web server and start the load. Write the time.

Expected: In alarm in about three minutes. An email. A third instance.

10

Terminate one instance from the console. Write the time, and again when its replacement is Healthy.

Expected: Draining, then a replacement nobody asked for.

```
$ for i in 1 2; do nohup timeout 600 yes \  
$      > /dev/null 2>&1 & done
```

Load both servers: the alarm reads the average.

Read the bill, then finish



- 11** Cost Explorer, from the first day of the module: group by Service, then by the `Owner` tag if it is offered.
Expected: The module-to-date figure, beside your ledger total.

- 12** Prove it, break it and tear down, in that order, before the break.
Expected: No load balancer when you leave the room.

Prove it, then break it



PROVE IT

- The DNS name serves from two instance ids
- A scale-out and a replacement in the activity history of `thara-web-asg`
- The email from `thara-alerts` arrived

BREAK IT

- Change the health check path of `thara-web-tg` to `/missing`
- Every target goes unhealthy. The page still loads
- The group replaces healthy servers. Revert quickly, and explain

Tear down before the break



TEARDOWN

- Delete `thara-web-asg`. Its instances go with it
- Delete `thara-alb`, then `thara-web-tg`. Write the time
- Delete the alarm. Stop `thara-bastion`
- Keep `thara-web-lt`, `thara-alerts`, `thara-alb-sg` and `thara-web-role`
- The load balancer existed for under two hours

Common mistakes



COMMON MISTAKE

Auto Scaling replaces the load balancer.

The ALB is separate and needs to be created.



COMMON MISTAKE

The ALB is in a subnet.

It spans subnets across AZs.

Before this afternoon



TASK

Your group

Bring the capstone diagram this afternoon, for the peer review.

FREE

Fault finder

Five hidden faults and the evidence for each. A warm-up for the drill.

Questions & Discussion



This afternoon: five faults, a review, and taking it all down.



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Revision 113852e